Chapter – 2: Exploring Substances: Acidic, Basic, Neutral

- 28th Feb. – school – science fair – celebrate National Science Day
- Entry gate – siblings – Ashwin, Keerthi – greeted with – white paper – very curious
- Little ahead – volunteer – spraying liquid on white sheet
- Siblings – got their sheets sprayed – surprise – Welcome to the Wonderful World of Science appeared on the paper – excited – **reason** behind it
- Curiosity – satisfied – Colourful World of Substances stall – many act. – colour changes – mixing substances

Nature – Our Science Laboratory

Litmus as an indicator

- Activity –
  - Collect samples – lemon juice, soap solu., amla juice, tamarind water, vinegar, baking soda solution, lime water, tap water, washing powder solution, sugar solution, salt solution
  - Take strip – blue litmus paper – cut – small pieces
  - Spread pieces – white tile
  - Use dropper – 1 drop – each sample – litmus paper
  - **Observe** – change in colour
  - **Record** – obs. in notebook
  - Repeat – same thing – red litmus paper
- Analyse – obs. – sort samples – 3 grp. –
  - Grp. A – turn blue to red
  - Grp. B – turn red to blue
  - Grp. C – do not affect any
- **Litmus** – natural substance – lichens – available – both forms – solution and paper strips – 2 colours – red, blue
- Substance – turn blue to red – **acidic** – turn red to blue – **basic**
- Litmus – diff. colour – acid, base solutions – also called – **acid-base indicator**
- Many other substances – natural, synthetic – used as indicators
- **Classify** – substances – grp. A, B, C –
  - Grp. A – lemon juice, amla juice, tamarind water, vinegar – turn blue to red – acidic
  - Grp. B – soap solution, baking soda solution, lime water, washing powder solution – turn red to blue – basic
  - Grp. C – tap water, sugar solution, salt solution – no change in colour – **neutral** – neither acidic nor basic
- Grp. A – all substances – edible – recall their taste – taste sour
- Any substance – taste sour – contains acids – acidic in nature
  - Lemon – citric acid
  - Amla – ascorbic acid (vitamin C)
  - Raw mango – tartaric acid
  - Tomato – oxalic acid
- **Caution** – do not taste any unknown substance – many can be dangerous
- Grp. B – take any substance – rub b/w fingers – feels soapy, slippery
- Any substance – feels soapy – taste bitter – contains base – basic in nature
- BUT – everything tasting bitter – not base – example – bitter gourd (*karela*)

Red rose as an indicator

- Activity –
    - Collect – fallen petals – red rose – do not pluck
    - Wash – fistful of petals with water
    - Crush petals – mortar, pestle – place in glass tumbler
    - Pour – hot water – glass tumbler – ensure – crushed petals – immersed
    - **Caution** – perform under adult supervision
    - Cover tumbler – lid – wait 5-10 minutes – water coloured – filter it
    - Filtrate – required extract – use as acid-base indicator
- Activity –
    - Place 10-20 drops – red rose extract – 2 test tubes – mark them A, B
    - Use droppers – 20-30 drops –
        - Lemon juice – test tube A
        - Soap solution – test tube B
    - Obs., record – colour changes
    - Repeat same – other samples – record obs.
- Discuss obs. – with classmates –
    - Samples – change colour – flower extract – shade of red – same – litmus – blue to red
    - Samples – change colour – flower extract – shade of green – same – litmus – red to blue
    - Samples – do not change colour – flower extract – same – litmus – no change
- Repeat process – prepare extract, test substances – vegetables, fruits, flowers – beetroot, purple cabbage, turmeric, blackberry, red hibiscus

Turmeric as an indicator

- Activity –
    - Take spoonful – turmeric (*haldi*) – petri dish – add little water – make paste
    - Carefully – dip a piece of filter paper in turmeric paste – gets yellow colour
    - Take out – let it dry
    - Cut yellow paper – thin strips – used as turmeric paper
    - **Caution** – perform under adult supervision
    - Use dropper – put drop – each sample – separate pieces – turmeric paper
    - Record obs.
- Grp. samples – do not change colour – turmeric paper
- **Compare** – samples – grp. A, B, C
- Turmeric paper – only used for basic substances – not for acidic, neutral substances
- Some substances – change odour – acidic, basic medium – called **olfactory indicators**
- Activity –
    - Take – finely chopped onion – along with strips – clean cloth
    - Tightly close container – leave it overnight – take 2 strips – check odour
    - Keep them – clean surface – few drops – tamarind water – one strip – baking soda solution – other strip
    - Check odour again – note obs. – test with other samples
- **Know a scientist –**
    - Acharya Prafulla Chandra Ray (P.C. Ray) – father of modern Indian Chemistry
    - Doctorate from U.K. – returned to India
    - 1901 – 1st Pharma Company – manufacturing medicines
    - Ray – also social reformer – advocated – use of mother tongue – medium of instruction

What Happens When Acidic Substances Mix with Basic Substances?

- Activity –
  - 1 drop lemon juice – test tube – add 20 drops – water – observe colour
  - Add – 1 drop – blue litmus solution – colour changes to red
  - Slowly – drop lime water – this test tube – use dropper – swirl it – obs. colour change
  - At a pt. – red colour – changes to blue
  - Add – 1 drop – lemon juice – colour changes again
- Blue litmus solution – added to lemon juice – turns red
- Adding lime water – colour changes to blue – indicating – solution – no longer acidic
- Lime water – neutralized – effects of acid
- Acidic solution – mix with basic solution – resulting solution – neither acidic nor basic – such reactions – **neutralisation**
- Neutralisation reaction – **salt**, water and heat evolved
  Acid + Base → Salt + Water + Heat

Neutralisation in Daily Life

- **Situation 1** –
  - Keerthi – obs. butterfly – garden – hand resting on tree
  - Suddenly – red ant bit her – skin red – stinging pain
  - Her brother – helped her – apply moist baking soda – relieves pain
  - Ant bite – inject formic acid – apply moist baking soda – basic – neutralize effects of acid
- **Situation 2** –
  - On farmer's online portal – 1 query – plants not growing well
  - Detailed discussion – found out – excessive use – chem. fertilisers – soil acidic
  - Soil too acidic – plants do not grow well – treat with lime – basic
  - Soil – basic – organic matter added – manure, compost leaves – release acids – neutralize effects of base
  - Sometimes – soil – neutral – yet – poor health – deficiency of nutrients
- **Situation 3** –
  - Ashwin's friend – Gurbir – stays near Industrial area
  - Fish population – neighbourhood lake – declining everyday
  - Reason – factory waste – released into lake
  - Factory waste – acidic – treat with bases – before releasing into water bodies